

Visual RAG for Complex PDFs

This document describes a multimodal retrieval-augmented generation system for long PDF documents. The main contribution is page-level grounding.

Motivation: complex PDFs mix text, tables, and figures. Pure OCR pipelines lose layout, while vision-only pipelines may ignore dense text.

Adaptive Hybrid Fusion

The system combines ColPali visual retrieval with BM25 text retrieval. Pages with high text density receive higher text-channel weight.

Fusion formula: $\text{fused} = w_v * s_v + w_t * s_t$, where w_t grows with normalized text density in $[0.25, 0.75]$.

Evaluation Results

On the sample benchmark, hybrid RAG achieved 87.3% answer accuracy. Refusal precision reached 90.1% on unanswerable questions.

Page recall@5 for hybrid retrieval was 0.84.

System Components

Component | Role

ColPali | Visual page retrieval

BM25 | Text page retrieval

Qwen2-VL | Multimodal answer generation

Verifier | Cite-then-Verify page check

The verifier asks YES/NO whether each cited page supports the claim.

Limitations and Scope

This report only evaluates English and Chinese technical PDFs under 50 pages. It does not discuss stock prices, celebrity biographies, or real-time news events. Training details for ColPali are omitted.